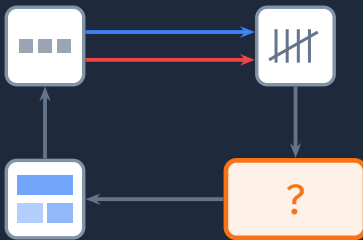


A Value Gate Pays Off When Habit Is Blind to Outcome

Evidence from a learner that builds a vocabulary
of reusable chunks to repair damaged strings



Jasper Gilley

The question

Repetition builds habits whether or not they pay

A goal-directed learner repeats actions, and repetition turns them into **habits**.

Repetition does not check outcomes: a practised error becomes a habit as readily as a success.

Animals also have an **evaluative system** that withholds reinforcement from repeated actions that do not pay.

The question

In a learner that builds skills by repetition, when does a value gate in front of memory pay off, and can the learner grow it from its own practice? Here, memory is a **phrasebook** of reusable chunks.



A learner that practises

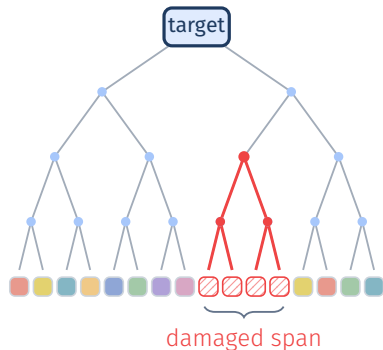
The task is to repair a damaged string so it derives its target again

A random **hierarchical grammar**: eight symbols per level, each with two rules that expand it into a pair one level down, six levels from root to 64 tokens. A level- ℓ node spans 2^ℓ tokens.

A problem is a valid string for a **target** root with the span under one node **rewritten**, so it no longer derives the target. The learner edits it; a **grader** passes any valid derivation of the target.

A goal, attempts, and a graded outcome: goal-directed behaviour in miniature, with exact ground truth.

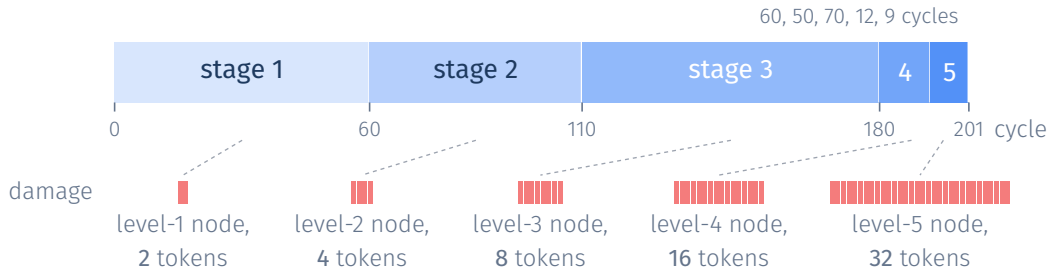
Grammar: the Random Hierarchy Model (Cagnetta et al., 2024).



4 of the 6 levels shown (16 of 64 tokens)

The curriculum demands higher-level chunks

The learner practises for **201 cycles** of 64 problems each. Stage k damages a level- k node, so the span to repair doubles: **2, 4, 8, 16, then 32 tokens**. Stage error is one minus that stage's held-out pass rate, averaged over its cycles.

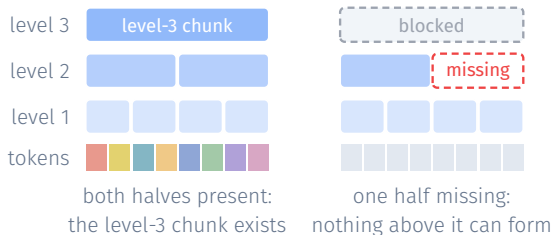


The last two stages give few cycles of evidence. Stage boundaries are the same at both seeds; when the learner may start using each level of chunks differs by seed.

Chunks make large repairs possible; a missing half blocks the levels above

Repairing 32 tokens one token at a time exceeds the search planner's budget. **Chunks** let it compose large repairs from a few reusable pieces.

The **phrasebook** is organised by level. The level-1 chunks, the eight bottom symbols, are given; each spans 2 tokens. Above that, every level- ℓ chunk is a pair of level- $(\ell-1)$ chunks and must be **earned**.



Nesting cuts both ways

A level-3 chunk requires both level-2 halves. A missing half **blocks** everything that would be built on it.

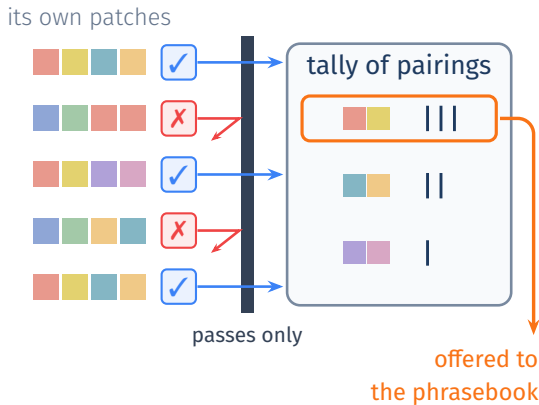
The habit counter grows the phrasebook by repetition

After each cycle the counter tallies pairings around each repair: the repaired level- k span and its neighbour form a candidate level- $(k+1)$ chunk. It works **one level ahead**.

A pairing seen **three times** is offered to the phrasebook, even before its level is usable: level-2 chunks enter in stage 1 and are first written at cycle 48 / 59.

By default

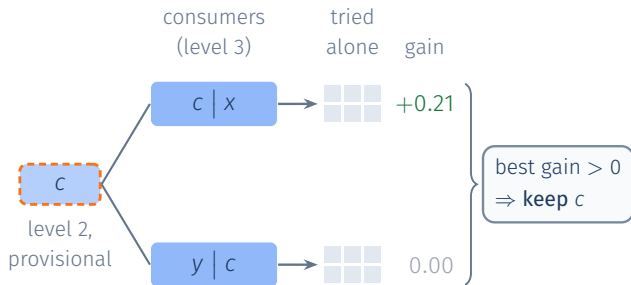
Only **successful** repairs count. Unlike habit, this counter checks outcomes.



The value gate admits chunks first, then tests what can be built on them

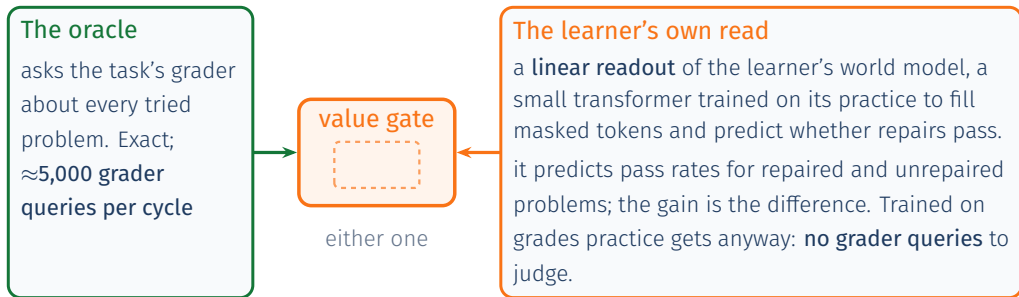
A chunk's worth must be tested, so every offered chunk enters **provisionally**. Its job is to be built on, so it is judged through its **consumers**: pairings one level up that contain it.

Each consumer is **tried alone** on 192 fresh problems damaged at its level, using only that chunk. Unrepaired, they pass at 0, so its gain is its pass rate.



- No consumer with positive gain: **remove**.
- No consumer within a timer: **remove** (a known flaw, kept).
- A new consumer enters the tally later: **offer again**.

Either the oracle or the learner's own read can judge chunks



The read waits for **512 of its own graded writes** at the consumers' level before judging.

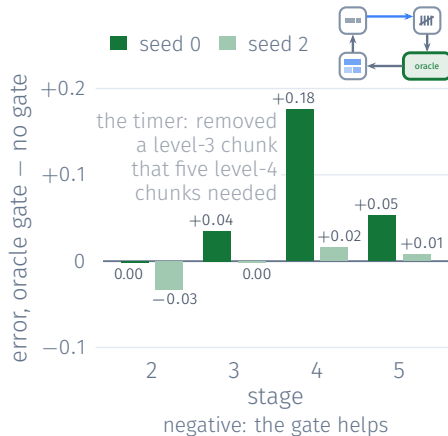
Two ways of counting

When only successes count, a value gate has nothing to add

With successes-only counting, even the **oracle** gate does no better than no gate, at either seed. What recurs in successes is mostly worth keeping: frequency is a close stand-in for value. The limit is how fast true chunks *arrive*, not the quality of what gets in.

Takeaway

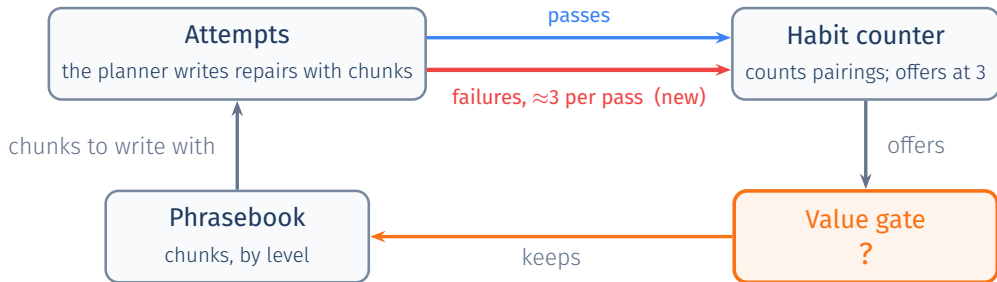
The success filter already does a value system's job. Real habit doesn't check outcomes, so no real learner has this filter built in.



stage 1 omitted: earned chunks are unusable for most of it, and all conditions agree within 0.005

One change blinds the counter to outcome: count the failures too

The counter still samples successful repairs exactly as before. It now also counts **failed repairs**, in the batch's own ratio: about **three failures for every success**.



Everything else is the same: grammar, planner, world model, and each seed's schedule of when chunk levels become usable. Gated runs branch from the ungated run at cycle 10 / 5, before any chunk is judged.

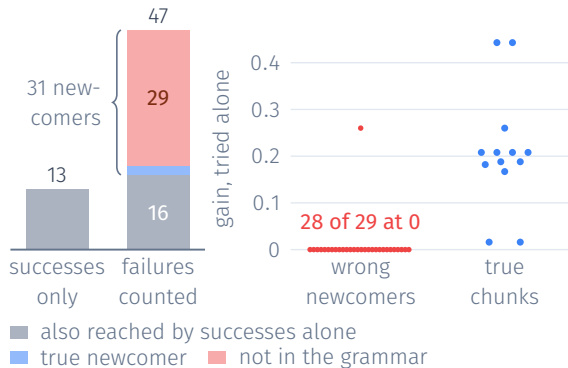
The junk is the learner's own recurring mistakes

The counter sees only the region around a repair, so a failure contributes **exactly its wrong patch**; at three failures per success, the most-repeated patches are errors.

By cycle 40 the level-2 counter holds 47 and 48 chunks (seeds 0 and 2), against 13 counting successes alone. Of the newcomers, 29 and 30 are not in the grammar; tried alone, **28 of 29** and **30 of 30** repair nothing. Added to the phrasebook so far, a quarter to a third make it worse.

True chunks are not delayed: all 14 the run finds arrive by cycle 50 / 25.

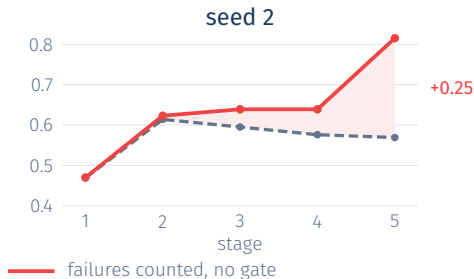
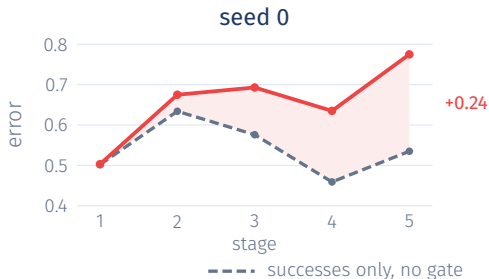
level-2 chunks offered
by cycle 40



seed 0, cycle 40. Labels compare spellings with the grammar's rules; a few legal chunks spell like illegal ones (one labelled false repairs 26%).

Without a gate, the learner memorises its mistakes and loses depth

The ungated phrasebook ends several times larger than the successes-only one, at about a third of its precision (the share labelled true): at level 2, **61 chunks at 0.23** against **17 at 0.65** (seed 0). As damage spans grow, error rises stage by stage above the successes-only learner.



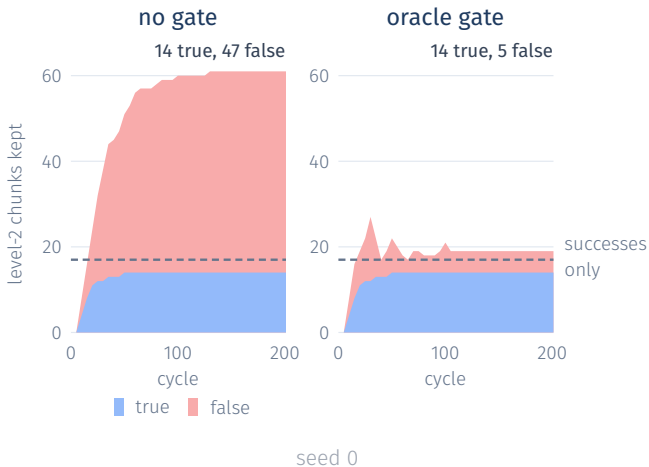
The value gate at work

The oracle gate rebuilds the phrasebook from the polluted stream

The **oracle** gate cuts the level-2 phrasebook to 17 chunks by cycle 40, the successes-only learner's final size. It ends with **14 true** and **5 false**, against 14 and 47 ungated (seed 0).

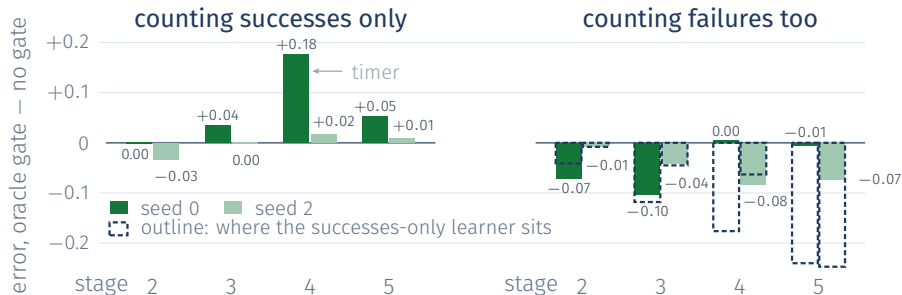
None of its **542 removals** (both seeds, by timer or by value) is of a chunk labelled true.

It beats the ungated run in 3 of 4 stages at seed 0 and 4 of 4 at seed 2, at $\approx 5,000$ grader queries per cycle.



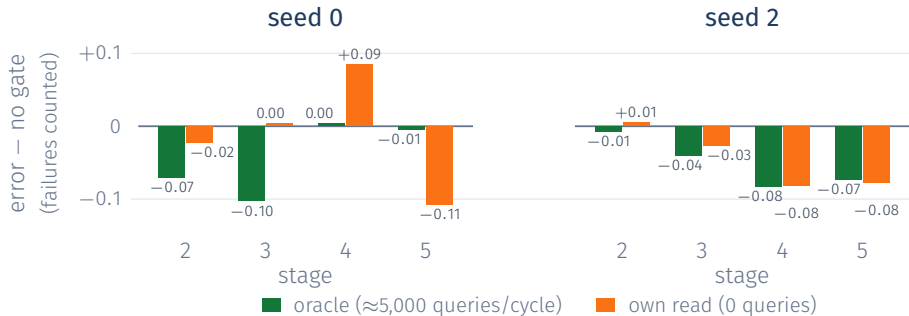
A value gate's payoff depends on what the counter filters out upstream

The oracle grades every trial exactly, so it shows whether the gate can help at all. Counting only successes, it adds nothing. When failures count, it closes the gap to the successes-only learner at stages 2–3 on both seeds and stage 4 at seed 2, but not at the last stage.



The learner's own read matches the oracle at one seed, with no grader queries

At seed 2 the **read** tracks the **oracle** stage for stage, within 0.01 at the last two stages, with **no grader queries** to judge chunks. At seed 0 its record is mixed: better than no gate at stages 2 and 5, worse at stage 4 (+0.09).

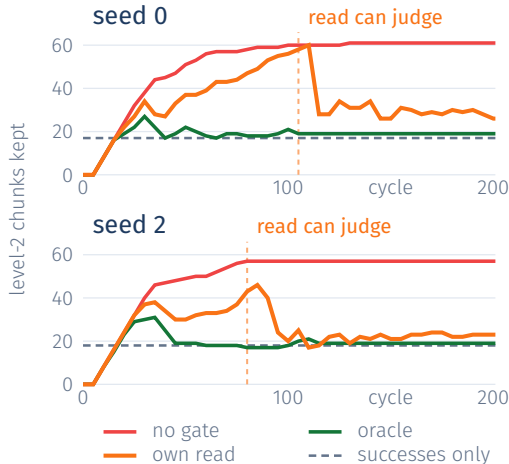


The read must wait for experience one level up

To judge a chunk, the read evaluates its consumers one level up and needs 512 of its own graded writes there. Level-3 chunks become usable only at cycle 100 / 77, so the read **lags by construction**: level-2 chunks are first judged at 105 / 80, level-3 at 197, level-4 never. Meanwhile junk enters unchecked.

Takeaway

No grader queries, but it costs time: blind habit keeps mistakes during the wait.



Once it can judge, the read catches about half of what the oracle catches

The read-gated runs also record the oracle's verdict, unused. Where the oracle would remove a level-2 chunk, the read removes it **42% / 45%** of the time; where it would keep one, the read removes it **10% / 18%**.

At a chunk's own level, with far more experience, the read ranks oracle-useful chunks above useless ones ($AUC \approx 0.75$), but its zero point is off: most true chunks get negative gains too.

		seed 0		seed 2	
		read removes	read keeps	read removes	read keeps
oracle	removes	127	173 missed junk	111	137 missed junk
	keeps	34 false alarm	313	72 false alarm	318

level-2 decisions, same tried problems

Proposed next step

Judge at the chunk's own level first, with a calibrated threshold, then through its consumers.

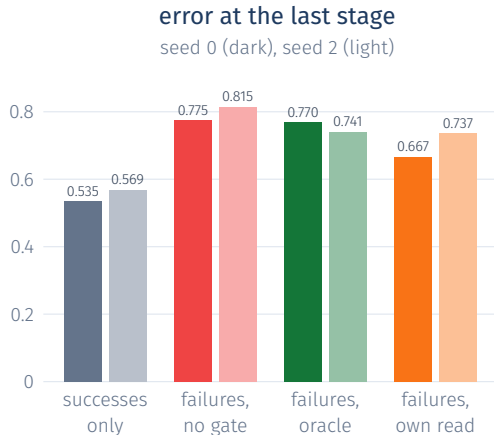
What it means

Neither gate fully recovers last-stage performance

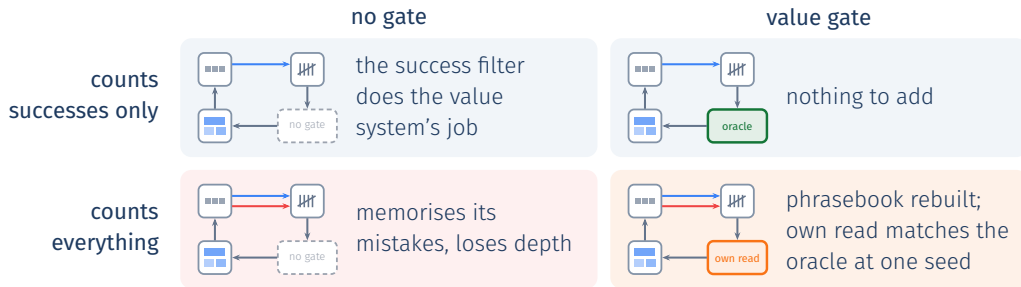
Both gates end **0.13–0.24** above the successes-only learner at the last stage, on both seeds. The data do not distinguish three possible causes:

- the top level is never judged, since nothing sits above it; its phrasebook stays two to three times the successes-only size;
- churn: about 270 removals and 150 re-offers per run, against 14 and 6 when only successes count;
- early practice used a polluted phrasebook.

These estimates rest on few cycles: the last two stages last 12 and 9 cycles, with one run per seed.



A value gate pays off when habit is blind to outcome



Goal-directed behaviour in the wild looks like the bottom row: candidates are abundant, habits form by repetition, and the hard part is knowing which to keep. **Habit and value need each other:** the habit loop's graded attempts train the read, and the read decides what the habit loop keeps.

A value gate pays off when habit is blind to outcome

Successes only

The success filter already does the value system's job; even the oracle adds nothing.

Failures counted

The learner memorises its mistakes and loses depth.

The oracle gate

Rebuilds the phrasebook; none of its 542 removals is labelled true.

The learner's own read

Matches the oracle at one seed with no grader queries, but must wait for experience one level up.